

Integration Plan

Three Follow-on Proposals, Two IEEE S&P Papers

RobustUAVs.ai research programme · Roger Nick Anaedevha · 2026-10-01

Supersedes the standalone framing of P1, P2 and P3; those files are kept unchanged as the record of the original proposals.

1 The decision

Two papers, not one and not three.

- **Paper I** (S&P main track) **merges P1 and P2**. *What Authenticity Costs in Metres: A Certified Budget for Cryptography and Sensing in UAV Navigation*.
- **Paper II** (S&P Systematization of Knowledge track) **is P3, broadened**. *SoK: The Quantifier Gap Between Certified Robustness and Airworthiness Assurance*.

The reason is a single observation that only becomes visible when P1 and P2 are read side by side. The parent manuscript's certificate is a budget,

$$\rho = \Delta \cdot \gamma \cdot e^{LT_c} \leq m, \quad (1)$$

and P1 and P2 each attack one of its factors. P1 says γ depends on which navigation source is trusted, and that *authenticating* a source is what makes it trustworthy. P2 says Δ does not care why an input is stale, and that *authenticating* a message makes it stale. The same operation therefore appears on both sides of Eq. (1) with opposite signs: authentication is simultaneously a credit (lower γ) and a debit (higher Δ). Neither proposal can say which wins. Together they can, and the answer is the paper.

2 Why not three papers

Each of P1 and P2 is weaker alone than its share of the merge, and each says so in its own risk section.

- **P1 alone is a measurement paper.** Its central claim is described in its own text as “arithmetic rather than new theory”; its contribution is the per-modality γ . That is a solid journal contribution (*TIFS*, *TCPS*, which is where P1 targeted it) and a weak S&P one.
- **P2 alone risks being read as post-quantum benchmarking.** Its risk section names exactly this: “reviewers may read it as post-quantum benchmarking”. Its mitigation, “lead with the budget result”, is only available if there is something for the budget to trade *against*. P1 supplies it.
- **Merged, there is a falsifiable result neither has:** a crossover inequality deciding when authentication increases the certified mission floor and when it decreases it (Paper I, Proposition 1). That is a finding about security engineering, not a measurement of a component.

3 Why not one paper

P3 does not merge, and forcing it in would damage both halves.

- **Different contribution type.** P1 and P2 produce measurements, a theorem corollary and an attack. P3 produces a mapping, a gap analysis and a generator. S&P main-track reviewers judge the first kind; the second kind is what the SoK track exists for.
- **Different audience.** P3's own framing: it “changes who the result is *for*”, from a security audience to a certification audience. A paper cannot change its audience halfway through.
- **No shared measurement.** P3 reuses the parent's committed results and needs nothing from P1 or P2. Bundling it would make Paper II's schedule hostage to Paper I's hardware campaign for no gain.

- **Page budget.** Paper I already carries two measurement campaigns in 13 pages. P3’s mapping tables alone would take three.

■ *P3 as written is not yet an S&P SoK. It maps one certificate (ours) onto standards. An SoK must systematise a literature. Paper II therefore broadens P3 from “our certificate as evidence” to “every class of certified guarantee for learned autonomy, organised by what it quantifies over” — see Paper II, §2. Without that broadening it should go to DASC, which P3 already named as a venue, rather than to S&P.*

4 Element-by-element categorisation

Every element of the three proposals, and where it goes. Legend: **I-core** = Paper I main body; **I-appx** = Paper I appendix; **II** = Paper II; **drop** = not carried forward.

4.1 From P1 (certified PNT substitution)

Element	Goes to	Why
Claim: γ is a property of the navigation source, not the attack	I-core	The “credit” side of the budget. Without it P2 has nothing to trade against.
Claim: authentication discharges the re-anchoring hypothesis	I-core	Splits Paper I’s analysis into two regimes (delay vs. persistent denial); in the denial regime it is what makes <i>any</i> certificate non-vacuous.
Claim: the certificate induces a sensor-selection rule	I-core (rule) I-appx (optimal-ity proof)	The rule is short and is the operator-facing output; its brute-force verification is evidence, its proof is length.
γ for GNSS	I-core	Done. Use 1.625 m/s (campaign supremum), not P1’s 1.195/1.365, which predate the interface campaign.
γ inertial-only	I-core	Data already on disk; about a week. Do first.
γ visual–inertial	I-core	Public VIO datasets; second measured modality beyond GNSS.
γ LEO signals of opportunity	I-appx	No receiver logs. A <code>physics_model</code> value from published RMSE is honest only if tagged and kept out of the headline.
Fusion model $\gamma_{\text{eff}} = \min_s \gamma_s$	I-core	Conservative and unglamorous, as P1 asked. Filter-weighted variant to the appendix.
Denial-duration sweep	I-core	Provides the regime-(ii) evidence.
Schema change (<code>pnt_source</code> , <code>pnt_authenticated</code>)	I-core (one paragraph)	Additive, keeps every existing record valid.
PNT Source Selector page	platform	Merged into one Paper I page; see §7.

4.2 From P2 (the security tax)

Element	Goes to	Why
Budget $\rho = (\Delta + \Delta_{\text{auth}})\gamma e^{LT_c} \leq m$	I-core	Becomes the merged budget, with $\gamma \rightarrow \gamma_{\text{eff}}(S, a)$.
Prop.: detector tightness buys cryptographic headroom	I-core	Kept, and sharpened: Paper I shows it holds for <i>affordability</i> while the opposite holds for <i>benefit</i> (Paper I, Proposition 2).
Prop.: a margin below which nothing certifies	I-core (already partly shown)	At the parent’s constants, $m = 2m$ certifies nothing <i>even with zero authentication cost</i> (the budget is -0.12 s). The proposition is partly a consequence of existing results; the new part is naming the margin per transport.
Prop.: ranking by size is the wrong ranking	I-core	The “cliff” result: fragmentation and bus occupancy, not per-message latency, bind.
Measure sign/verify latency on MCU and companion board	I-core	The measurement that keeps the result from restating its inputs.
DroneCAN fragmentation model	I-core	Validated against benign HCRL timing.
Induced-verification attack	I-core	The S&P hook. Must be positioned against TESLA-era prior art on signature-flooding DoS [1]: the mechanism is known, the mission-level bound is new.
TESLA-style delayed authentication as a scheme	I-core (new)	Not in P2. It is a scheme whose Δ_{auth} is an explicit, tunable parameter — the cleanest possible test of the budget.
Candidate scheme set pinned to FIPS 203/204/205	I-core	As P2 said: lets a future scheme be placed without re-running.
Crypto Budget page	platform	Merged into one Paper I page; see §7.
pq_swarm_kyber (MDPI)	drop	Key agreement, not per-message authentication; not load-bearing for Paper I, and the supervisor’s guidance is to avoid MDPI unless core.

4.3 From P3 (certificate as compliance artifact)

Element	Goes to	Why
“Certified radii are per-input; assurance objectives are per-operation”	II (central axis)	This sentence, buried in P3’s honesty box, is the organising idea of the SoK. Promoted to the title.
Vocabulary mapping (radius, tube, operating region $\leftrightarrow$ ODD, contingency volume)	II	One row per certificate class, not just ours.
Discharge / evidence / untouched gap analysis	II	Applied to every surveyed guarantee class, not only the four in the parent paper.
Simplex runtime assurance as the bridge	II	Developed as P3 asked.
SORA Annex A evidence generator	II (artifact)	What makes the SoK more than a survey.
Compliance Pack export	platform	One export action.

5 Categorisation by contribution type

The same elements, grouped by the kind of claim a reviewer will judge them as.

Type	Paper I	Paper II
Theory	Merged budget; crossover inequality; two-regime split (delay vs. denial); selection rule	Quantifier taxonomy (per-input, per-trajectory, per-mission, per-population)
Measurement	γ for GNSS, inertial, VIO; Δ_{auth} for HMAC, Ed25519, ML-DSA, SLH-DSA, TESLA; bus occupancy	none (reuses committed results)
Attack	Induced verification, with bound and mitigation	none
Systematisation	—	Literature of certified guarantees for learned autonomy, mapped onto SORA / ED-324 / EASA learning assurance
Artifact	Budget calculator, crossover map, selection rule	SORA Annex A evidence generator

6 Which of the parent’s limitations each paper closes

Parent limitation	Paper I	Paper II	Note
Re-anchoring is assumed	closes	—	Authentication makes the hypothesis satisfiable; denial regime analysed explicitly.
γ calibrated on one platform, one regime	partly	—	Adds modalities; still <i>not</i> a cruise campaign. That gap remains data-gated (PENDING_ON_DATA.md).
One attack surface (mesh delay)	partly	—	Adds the bus (intra_bus) and C2 link (c2_link) on the defence side.
Guarantee bounds a population, not a flight	—	reframes	The SoK’s central finding is that this is a general property of the field, not a weakness of one paper.
Evidence-class imbalance	no	no	Unchanged by either. Say so.

7 Platform integration

Three proposed pages collapse to two.

- **Certified Budget** (Paper I): replaces both the *PNT Source Selector* and the *Crypto Budget* pages. Choose a sensor suite with authentication states, a transport, a scheme, a message rate and a margin; see the budget, the crossover sign, and the bus occupancy, live. One endpoint, `/api/certify/budget`.
- **Compliance Pack** (Paper II): an export action, unchanged from P3.

8 Readiness against the S&P bar

Reviewer will ask	Paper I	Paper II
Is there a new result, not just a measurement?	Yes: the crossover and the cliff.	Yes, if broadened: the quantifier gap as a field-wide finding.
Is the threat model explicit?	Inherited from parent; adds the induced-verification adversary.	Not applicable; must state its inclusion criteria instead.
Are numbers measured or modelled?	Mixed; every number carries a provenance tag. LEO is modelled.	Reuses measured results.
Is there an attack?	Yes, with a bound and a mitigation.	No, and an SoK does not need one.
Biggest risk	Hardware access for the MCU benchmarks.	A thin literature corpus. Needs a systematic search with stated criteria (§2 of Paper II).
Ready to start	Now: inertial γ and the budget arithmetic need nothing new.	Now: needs no measurement.

9 Sequencing

1. **Week 1–2, Paper I:** inertial-only γ from the Whelan logs already on disk. This is also the measurement that most strengthens the *parent* manuscript, whose cruise discussion currently asserts this value from an error budget.
2. **Week 1–6, Paper II:** systematic literature search; this runs in parallel and depends on nothing.
3. **Week 3–10, Paper I:** MCU and companion-board benchmarks; VIO γ ; fragmentation validation against HCRL.
4. **Week 8–12, Paper I:** induced-verification attack and mitigation.
5. **Submit Paper II first.** It is cheaper and lower-risk, and an SoK that cites the parent establishes the vocabulary Paper I then uses.

■ *Paper I carries the larger risk, because it has two measurement campaigns and either can fail. If the MCU benchmarks cannot be run on real flight-controller hardware, the fallback is the pinned-ARM measurement P2 already named, tagged as such — and the crossover result survives, because it depends on the ratio $\Delta_{\text{auth}}/\Delta(\theta)$, which a pinned ARM target still bounds. If the γ campaign yields only GNSS and inertial, two modalities still establish that γ is source-dependent. The paper degrades gracefully; it does not collapse to P1 or P2 alone.*

References

- [1] Adrian Perrig, Ran Canetti, J. D. Tygar, and Dawn Song. Efficient Authentication and Signing of Multicast Streams over Lossy Channels. In *Proc. IEEE Symposium on Security and Privacy (S&P)*, 2000.